

Smart Autonomous Pantry Replenishment Device

Project Overview

The objective of this project is to design and develop a standalone AI-powered device that autonomously monitors pantry inventory and triggers grocery replenishment workflows using only a camera feed as input.

The system observes a predefined dry pantry/storage area through a mounted camera, identifies items, estimates when inventory is running low, and automatically places or recommends orders through platforms such as Blinkit, Zepto, or Swiggy Instamart.

The project is divided into two independent but complementary problem statements. Applicants may opt for either track based on their interest and skillset.

PS-1 : AI, Software & Embedded Systems

Objective

Build an end-to-end vision-driven pipeline that detects pantry items, tracks inventory in real time, estimates low-stock conditions, runs efficiently as edge-AI inference, and autonomously triggers grocery reorders - all from a single camera feed.

Core Objectives

- Continuous pantry monitoring through a live camera feed
- Automatic item detection, classification, and tracking
- Visual estimation of remaining quantity and low-stock conditions
- Consumption-aware, automated reorder triggering
- Real-time edge-AI inference on a resource-constrained device

Technical Challenges

- Reliable item detection in cluttered, dynamic scenes with varying lighting, occlusion, and changing layouts.
- Estimating remaining quantity from visual cues alone across diverse packaging formats - packets, jars, boxes, and containers.
- Optimizing models for real-time inference, low-power operation, and stable connectivity on edge hardware.
- Building a replenishment pipeline that maps detected items to grocery-platform catalogues and handles thresholds, quantity estimation, and ordering workflows.

Expected Deliverables

A functional CV pipeline for detection, tracking, and low-stock estimation; an edge-AI inference setup; and an automated reorder module integrated with at least one grocery platform.

PS-2 : CAD, Hardware & Device Design

Objective

Design and development of a compact standalone hardware device, including enclosure design, camera-mounting architecture, modular deployment system, manufacturable CAD models, and embedded hardware integration.

Core Objectives

- Compact, deployable enclosure for the edge-AI compute unit
- Stable camera-mounting architecture with adjustable field of view
- Modular deployment system suitable for varied pantry layouts
- Manufacturable CAD models ready for prototyping
- Clean integration of embedded electronics, power, and connectivity

Technical Challenges

- Creating a mounting architecture that maintains a reliable camera viewpoint across different storage geometries.
- Ensuring modularity so the device can be deployed across multiple pantry/storage layouts with minimal rework.
- Integrating the camera, compute board, power, and connectivity within a manufacturable form factor.

Expected Deliverables

Complete CAD assemblies (enclosure + mount), manufacturing-ready drawings, a physical prototype of the enclosure with integrated electronics, and documentation of the deployment system.

Final Outcome

A functional standalone pantry-monitoring prototype that detects and tracks items via camera input, estimates low-stock conditions, and triggers automated reorder workflows - housed in a deployable, manufacturable enclosure built around an embedded edge-AI hardware setup.